Appendix A — Program

Source: Project Renegade Living + Lodging Prototype, cover sheet (06/20/25). Areas are the total for each row, as in the source schedule.

Count	Type	Total area (SF)
Shared		
1	Corridor	1,335
1	Gym	968
1	Lobby	2,865
1	Mens	205
1	Womens	200
5		5,573
BOH		
1	BOH	69
1	Hotel Kitchen / Pantry	331
1	Housekeeping	103
5	Ice	297
5	IT	372
5	Jan	269
1	Janitor	107
1	Loading	282
1	MEP Service	309
12	Stair	1,868
5	Supplies	394
38		4,401
Tenant		
1	White Box Retail	3,533
1		3,533
Hotel Units		
1	ADA Corner Suite	852
2	ADA King Spa	920
2	ADA QQ	892
2	ADA Suite	1,278
2	Corner Suite	1,728
69	King	23,692
3	King Spa	1,379
33	QQ	11,342
3	Suite	1,920
117		44,003
Business		
1	BOH / Bus	1,718
1	Luggage	122
2		1,840
163	**Total**	**59,350**

Appendix B — Gross area

Level	Source sheets (SF)	This schematic (SF)	Keys
Level 1	12,485	13,575	0
Level 2	12,342	12,587	24
Level 3	12,342	12,587	24
Level 4	12,342	12,587	23
Level 5	12,342	12,587	23
Level 6	12,342	12,587	23
Total	74,195	76,508	117

Meeting rule: 59,350 SF net $\times$ 1.33 $\times$ 1.08 = 85,300 SF (rounded up to the nearest 100 SF). The source sheets carry 74,195 SF, a ratio of 1.25. The two rules disagree by 11,105 SF; see open question 3.

Appendix C — Code checks

Screened against RNGD's 2021 IBC rule set. Planning level only, not a code determination.

Rule	Check	Required	Provided	Status
-	Keys placed	117	117	PASS
IBC-R1-007	Stories above grade	$\leq$ 5 (Type IIA, sprinklered R-1)	6	FAIL
IBC-R1-005	Building height (ft)	$\leq$ 85 (Type IIA)	69.8	PASS
IBC-R1-010	Area per story (SF)	$\leq$ 72,000 (Type IIA)	13,575	PASS
IBC-R1-012	High-rise screen (ft to top floor)	$\leq$ 75	59.0	PASS
IBC-R1-015	Exits per story	2 for 1-500 occupants	2 stairs; max load 316	PASS
IBC-R1-018	Exit separation (ft)	$\geq$ 1/3 diagonal = 75	207	PASS
IBC-R1-023	Exit access travel (ft)	$\leq$ 250 (sprinklered)	130	PASS
IBC-R1-025	Corridor width (in)	$\geq$ 44	72	PASS
IBC-R1-027	Dead-end corridor (ft)	$\leq$ 50 (sprinklered R-1)	5	PASS
IBC-R1-028	Egress door clear width (in)	$\geq$ 32	34 (3'-0" leaf)	PASS
IBC-R1-003	Automatic sprinklers	NFPA 13 required	assumed	ASSUMED
ADA-224.2	Accessible guest rooms (mobility)	$\geq$ 7 for 117 rooms (5 + 2 with roll-in)	7	PASS
ADA-224.4	Rooms with communication features	$\geq$ 12	not designated	REVIEW
ADA-224.5	Dispersion across room types	spread over 3 types	3 types: king, suite, two-bed	PASS
-	Fits buildable envelope	170 x 265 ft	234 x 58 ft	PASS

- IBC-R1-007: Level 1 counts as a story above grade plane. A Type IA podium (IBC 510.2) could allow 5 stories above it.
- IBC-R1-023: Half the stair-to-stair run plus the room depth.
- ADA-224.2: Confirm which 2 of the ADA units get roll-in showers (Table 224.2).
- ADA-224.4: Designate 12 rooms; at most 10% of the mobility rooms may count (224.5).

Appendix D — RNGD IBC rule set

42 rules from the CSV. 11 are evaluated against this schematic (column “Checked”). Most of the rest need space-level occupancy or detailed egress modelling.

Rule	Category	Rule	Op	Value	Units	Hard	Checked
IBC-R1-001	Occupancy	R-1 transient hotel classification	eq	R-1	classification	Hard	no
IBC-R1-002	Mixed Use	Small accessory assembly space classification	<	50 occupants or 750 sf	threshold	Hard	no
IBC-R1-003	Fire Protection	Automatic sprinkler required for Group R	requires	Automatic sprinkler system	system	Hard	yes
IBC-R1-004	Height / Stories	Allowable height and stories depend on occupancy, construction type, and spri...	lookup	Table 504.3 / 504.4	lookup	Hard	no
IBC-R1-005	Height / Stories	R occupancy sprinklered Type IIA allowable height	≤	85	ft	Hard	yes
IBC-R1-006	Height / Stories	R occupancy sprinklered Type IIB allowable height	≤	75	ft	Hard	no
IBC-R1-007	Height / Stories	R-1 sprinklered Type IIA allowable stories	≤	5	stories above grade plane	Hard	yes
IBC-R1-008	Height / Stories	R-1 sprinklered Type IIB allowable stories	≤	5	stories above grade plane	Hard	no
IBC-R1-010	Area	R-1 Type IIA allowable area factor, multistory sprinklered	≤	72000	sf per story base factor	Hard	yes
IBC-R1-011	Area	R-1 Type IIB allowable area factor, multistory sprinklered	≤	48000	sf per story base factor	Hard	no
IBC-R1-012	High-Rise Screen	High-rise threshold screen	>	75	ft to occupied floor	Hard	yes
IBC-R1-013	Egress	R-1 single-exit space maximum occupant load	≤	10	occupants	Hard	no
IBC-R1-014	Egress	R-1 single-exit common path maximum	≤	75	ft	Hard	no
IBC-R1-015	Egress	Minimum exits per story: 1-500 occupants	eq	2	exits/access to exits	Hard	yes
IBC-R1-016	Egress	Minimum exits per story: 501-1000 occupants	eq	3	exits/access to exits	Hard	no
IBC-R1-017	Egress	Minimum exits per story: over 1000 occupants	eq	4	exits/access to exits	Hard	no
IBC-R1-018	Egress	Exit separation in sprinklered building	≥	1/3 of maximum overall diagonal	ratio	Hard	yes
IBC-R1-019	Egress	Stair minimum width	≥	44	in	Hard	no

Rule	Category	Rule	Op	Value	Units	Hard	Checked
IBC-R1-020	Egress	Stair minimum width for occupant load under 50	≥	36	in	Hard	no
IBC-R1-021	Egress	Stair maximum riser height	≤	7	in	Hard	no
IBC-R1-022	Egress	Stair minimum tread depth	≥	11	in	Hard	no
IBC-R1-023	Egress	R exit access travel distance, sprinklered	≤	250	ft	Hard	yes
IBC-R1-024	Corridors	R corridor fire-resistance rating	eq	0.5	hour	Hard	no
IBC-R1-025	Corridors	Default minimum corridor width	≥	44	in	Hard	yes
IBC-R1-026	Corridors	Minimum corridor width when occupant load is under 50	≥	36	in	Hard	no
IBC-R1-027	Corridors	Dead-end corridor maximum for sprinklered R-1	≤	50	ft	Hard	yes
IBC-R1-028	Doors	Minimum clear width of required egress door opening	≥	32	in clear	Hard	yes
IBC-R1-029	Egress	Minimum means-of-egress ceiling height	≥	7.5	ft	Hard	no
IBC-OL-001	Occupant Load	Area-based occupant load calculation	formula	Area ÷ factor	occupants	Hard	no
IBC-OL-002	Occupant Load	Assembly without fixed seats - concentrated	divide_round_up	7	sf/person net	Hard	no
IBC-OL-003	Occupant Load	Assembly without fixed seats - standing	divide_round_up	5	sf/person net	Hard	no
IBC-OL-004	Occupant Load	Assembly without fixed seats - unconcentrated	divide_round_up	15	sf/person net	Hard	no
IBC-OL-005	Occupant Load	Business areas	divide_round_up	150	sf/person gross	Hard	no
IBC-OL-006	Occupant Load	Exercise rooms	divide_round_up	50	sf/person gross	Hard	no
IBC-OL-007	Occupant Load	Kitchens, commercial	divide_round_up	200	sf/person gross	Hard	no
IBC-OL-008	Occupant Load	Accessory storage / mechanical equipment room	divide_round_up	300	sf/person gross	Hard	no
IBC-OL-009	Occupant Load	Mercantile	divide_round_up	60	sf/person gross	Hard	no
IBC-OL-010	Occupant Load	Mercantile storage / stock / shipping	divide_round_up	300	sf/person gross	Hard	no
IBC-OL-011	Occupant Load	Parking garages	divide_round_up	200	sf/person gross	Hard	no
IBC-OL-012	Occupant Load	Residential	divide_round_up	200	sf/person gross	Hard	no
IBC-OL-013	Occupant Load	Pool / rink area	divide_round_up	50	sf/person gross	Hard	no
IBC-OL-014	Occupant Load	Pool / rink decks	divide_round_up	15	sf/person gross	Hard	no

Appendix E — RNGD workbook

Workbook sheet: Project_Config

Field	Working Value	Status / Note
Base code	2021 International Building Code	Fixed

Field	Working Value	Status / Note
Jurisdiction layer		Per RNGD direction; base code only
Project type	New construction	Fixed for prototype
Primary occupancy	Group R-1 transient hotel	Fixed
Stories	5 stories above grade plane	Fixed for current prototype
Sprinkler protection	Fully sprinklered; NFPA 13 assumption	Fixed for current prototype
Construction type	Type II; IIA vs IIB not yet locked	Keep both applicable Chapter 5 rows until subtype is confirmed
High-rise strategy	Screen only	Five-story prototype should still be checked against the 75-ft occu...
Accessibility scope	Not yet decided	IBC Chapter 11 can be added later if the team wants it inside this ...

Workbook sheet: Occupant_Load_Factors

factor_id	IBC function of space	factor	area_basis	machine_key
OL-F01	Assembly without fixed seats - concentrated	7	net	assembly_concentrated
OL-F02	Assembly without fixed seats - standing	5	net	assembly_standing
OL-F03	Assembly without fixed seats - unconcentrated	15	net	assembly_unconcentrated
OL-F04	Business areas	150	gross	business
OL-F05	Exercise rooms	50	gross	exercise_room
OL-F06	Kitchens, commercial	200	gross	commercial_kitchen
OL-F07	Accessory storage / mechanical equipment room	300	gross	accessory_storage_mechanical
OL-F08	Mercantile	60	gross	mercantile
OL-F09	Mercantile storage / stock / shipping	300	gross	mercantile_storage
OL-F10	Parking garages	200	gross	parking_garage
OL-F11	Residential	200	gross	residential
OL-F12	Pool / rink	50	gross	pool_rink
OL-F13	Pool / rink decks	15	gross	pool_deck

Workbook sheet: Hotel_Space_Mapping

reference_space	reference_area_sf	suggested_machine_key	factor	area_basis
Lobby	2865	assembly_unconcentrated	15	net
Gym	968	exercise_room	50	gross
Hotel Kitchen / Pantry	331	commercial_kitchen	200	gross
MEP Service	309	accessory_storage_mechanical	300	gross
White Box Retail	3533	mercantile	60	gross
Housekeeping	103	accessory_storage_mechanical	300	gross
Supplies	79	accessory_storage_mechanical	300	gross

reference_space	reference_area_sf	suggested_machine_key	factor	area_basis
Janitor	54	accessory_storage_mechanical	300	gross
King guestroom	344	residential	200	gross
QQ guestroom	344	residential	200	gross

Workbook sheet: Rule_Test_Cases

test_id	test_type	rule_id	scenario	area_sf
TC-001	occupant_load_calc	IBC-OL-012	344 SF King guestroom mapped to Residential	344.0
TC-002	occupant_load_calc	IBC-OL-006	968 SF hotel gym	968.0
TC-003	occupant_load_calc	IBC-OL-007	331 SF commercial kitchen/pantry	331.0
TC-004	occupant_load_calc	IBC-OL-009	3,533 SF white-box retail	3533.0
TC-005	occupant_load_calc	IBC-OL-004	2,865 SF lobby provisionally mapped to unconcentrated assembly	2865.0
TC-006	compliance	IBC-R1-015	Story occupant load 191 with 2 exits	
TC-007	compliance	IBC-R1-016	Story occupant load 600 with only 2 exits	
TC-008	compliance	IBC-R1-023	Sprinklered Group R travel distance of 240 ft	
TC-009	compliance	IBC-R1-023	Sprinklered Group R travel distance of 260 ft	
TC-010	compliance	IBC-R1-027	Sprinklered R-1 dead-end corridor of 55 ft	

Appendix F — Assumptions

- Envelope from the app sidebar (default synthetic lot 170 x 265 ft). Street assumed on the south side.
- Double-loaded corridor, 6 ft clear; room depth 26 ft; room width = program area / 26 ft.
- Floor-to-floor: Level 1 16 ft, guest floors 10.75 ft (from the L+L elevations).
- Core at the west end (Stair 1, two elevators, shafts); Stair 2 at the east end of the north side.
- Rows whose count equals the number of guest floors (Ice, IT, Jan, Supplies) are placed once per floor.
- Room mix spread evenly across floors, largest units first. Fixtures and furniture are typical, not from RNGD unit drawings.
- Occupant loads use the RNGD CSV factors (IBC Table 1004.5); lobby as assembly unconcentrated (15 SF net).
- 3D views are drawn with a painter's-algorithm axonometric in matplotlib. OpenGL renderers (moderngl, pyvista) are not installed in this environment.

Appendix G — Open questions for RNGD

1. Which area of the Level 1 / Level 2 drawing was meant? The reference coordinate note (scale 1.08 to the original image) did not name a room, stair or gridline. Please mark the spot on the L+L plans sheet.
2. Story limit: the program needs Level 1 plus five guest floors (six stories), but RNGD rule IBC-R1-007 allows five stories for sprinklered Type IIA R-1. Is a Type IA podium (IBC 510.2) intended, as in the L+L prototype?
3. Confirm the gross-up rule: kickoff meeting said 33% circulation + 8% wall burden; the L+L source sheets imply 1.25.
4. Real lot, setbacks, street frontage and fire-lane requirements (the envelope used here is synthetic).
5. Is parking part of this phase? It is not modelled.
6. Confirm the per-floor support rooms (Ice, IT, Jan, Supplies) and the Level 1 BOH arrangement.
7. Column grid: 30-34 ft bays assumed from the kickoff meeting. Is there a preferred structural module?
8. Prefab panel / module catalog to replace the indicative panel breakdown on the elevations.

Appendix H — Sources

- Project Renegade Living + Lodging Prototype drawings, cover sheet space allocation and plans (06/20/25).
- RNGD Model Training data: ADA Suite, Corner Suite and King groups (floor plans and 3D views); Schemes 1-3.
- RNGD IBC 2021 R1 rule set: RNGD_IBC_2021_R1_Five_Story_AI_Rules_v0_2.csv and the companion xlsx workbook.
- Kickoff meeting transcript (circulation and wall-burden rule, 30-34 ft bays, 2D-first output, upload of programs).
- RNGD Product Requirements Document (hotels as the first use case; proposal contents).